Results pack Q1: Positive fluid balance and 28-day mortality

FLUID-ICU is a SIMULATED dataset made for teaching. No real patients. Neudata | Foundations of Clinical Research, Level 1

SIMULATED DATA: NOT REAL PATIENTS. FLUID-ICU was generated by computer for this course. Write your manuscript as a training exercise and state clearly that the data are simulated.

1. The question

In adults with sepsis in the ICU, is a positive cumulative fluid balance by day 3 associated with higher 28-day mortality?

2. Design and data

- Design: multicentre prospective cohort study (observational).
- Setting: 4 ICUs (ICU A-D), consecutive adults with sepsis, enrolment January 2023 to December 2024.
- Exposure: positive cumulative fluid balance at day 3 (fluid in minus fluid out over ICU days 1-3 > 0 ml/kg) vs negative or zero balance.
- Outcome: death from any cause within 28 days of ICU admission (died_28d); time to death for survival analysis (time_to_event_28d, censored at day 28).
- Analysis sample: 618 of 620 patients (2 without a day-3 balance). Adjusted models: 527 complete cases.
- Pre-specified confounders (SAP): age, SOFA score, lactate, vasopressor, mechanical ventilation, chronic kidney disease.
- Software: R 4.6 (packages gtsummary, survival, mice). Scripts: Course/Data/R/.

Participant flow: 620 patients enrolled -> 2 without a day-3 fluid balance (weight or output missing after cleaning) -> 618 in the exposure analysis -> 91 without lactate (or age) -> 527 in the complete-case adjusted model.

3. Table 1: who was studied

Table 1. Baseline characteristics by fluid-balance group

Characteristic	Overall N = 618[1]	Negative balance N = 245[1]	Positive balance N = 373[1]
Age, years	62.6 (15.2)	64.6 (15.0)	61.4 (15.2)
Missing	1	1	0
Sex			
Female	243 (39%)	102 (42%)	141 (38%)
Male	375 (61%)	143 (58%)	232 (62%)
Weight, kg	64.7 (11.9)	65.1 (12.4)	64.4 (11.6)
Diabetes	158 (26%)	70 (29%)	88 (24%)
Chronic kidney disease	64 (10%)	28 (11%)	36 (9.7%)
Heart failure	97 (16%)	54 (22%)	43 (12%)
Hypertension	282 (46%)	124 (51%)	158 (42%)
Source of infection			

Characteristic	Overall N = 618[1]	Negative balance N = 245[1]	Positive balance N = 373[1]
Respiratory	267 (43%)	101 (41%)	166 (45%)
Urinary	126 (20%)	54 (22%)	72 (19%)
Abdominal	106 (17%)	46 (19%)	60 (16%)
Skin/soft tissue	57 (9.2%)	18 (7.3%)	39 (10%)
Other	62 (10%)	26 (11%)	36 (9.7%)
SOFA score	7 (5, 9)	6 (4, 8)	8 (6, 10)
APACHE II score	19 (15, 24)	17 (12, 21)	20 (17, 25)
Missing	6	3	3
Lactate, mmol/L	2.6 (1.7, 3.7)	2.2 (1.4, 3.0)	2.9 (1.9, 4.0)
Missing	91	50	41
Mean arterial pressure, mmHg	73 (66, 80)	76 (70, 84)	71 (65, 78)
Missing	4	1	3
Vasopressor on admission	266 (43%)	61 (25%)	205 (55%)
Mechanical ventilation	313 (51%)	97 (40%)	216 (58%)

[1]Mean (SD); n (%); Median (Q1, Q3)

Mean (SD); n (%); median (Q1, Q3). Missing = number of patients without that value. Table 1 describes the patients at ICU admission only. The outcomes, such as 28-day death, are in the main results.

Read Table 1 first: the positive-balance group was **sicker** at baseline: median SOFA 8 (6 to 10) vs 6 (4 to 8), vasopressor 205/373 (55.0%) vs 61/245 (24.9%). This is confounding by indication: sicker patients receive more fluid AND die more.

4. Main results

Table 2. 28-day mortality by fluid-balance group (unadjusted)

Outcome	Negative balance	Positive balance	Effect (95% CI)
Died by day 28, n/N (%)	45/245 (18.4%)	134/373 (35.9%)	Risk difference 17.6 (95% CI 10.7 to 24.4), RR 1.96, chi-square p <0.001
Survival at day 28 (Kaplan-Meier), %	81.6	64.1	Log-rank p <0.001

Table 3. Logistic regression for 28-day death: crude and adjusted odds ratios (adjusted model n = 527)

Variable	Crude OR (95% CI)	Crude p	Adjusted OR (95% CI)	Adjusted p
Positive fluid balance (vs negative)	2.49 (1.70 to 3.70)	<0.001	1.62 (1.02 to 2.59)	0.044
Age, per year	1.02 (1.01 to 1.03)	<0.001	1.02 (1.01 to 1.04)	0.002
SOFA score, per point	1.26 (1.18 to 1.35)	<0.001	1.10 (1.01 to 1.21)	0.039

Variable	Crude OR (95% CI)	Crude p	Adjusted OR (95% CI)	Adjusted p
Lactate, per mmol/L	1.38 (1.23 to 1.55)	<0.001	1.15 (1.00 to 1.32)	0.059
Vasopressor	3.31 (2.31 to 4.78)	<0.001	1.86 (1.20 to 2.88)	0.005
Mechanical ventilation	2.98 (2.07 to 4.34)	<0.001	1.69 (1.11 to 2.60)	0.016
Chronic kidney disease	2.58 (1.52 to 4.36)	<0.001	1.85 (1.00 to 3.44)	0.050

- Crude OR (all 618 patients): **2.49 (95% CI 1.70 to 3.70)**, p <0.001.
- Crude OR in the same 527 complete cases: 2.22 (95% CI 1.48 to 3.39).
- Adjusted OR (age, SOFA, lactate, vasopressor, ventilation, CKD): **1.62 (95% CI 1.02 to 2.59)**, p 0.044.
- Kaplan-Meier survival at day 28: 81.6% (negative) vs 64.1% (positive); log-rank p <0.001.
- Cox hazard ratio: crude 2.17 (95% CI 1.55 to 3.05); adjusted **1.46 (95% CI 1.00 to 2.13)**, p 0.053 (proportional-hazards test, global p = 0.10).

Table 4. Cox regression for time to death within 28 days

Model	n	Events	HR (95% CI)	p
Crude	618	179	2.17 (1.55 to 3.05)	<0.001
Adjusted (pre-specified confounders)	527	161	1.46 (1.00 to 2.13)	0.053

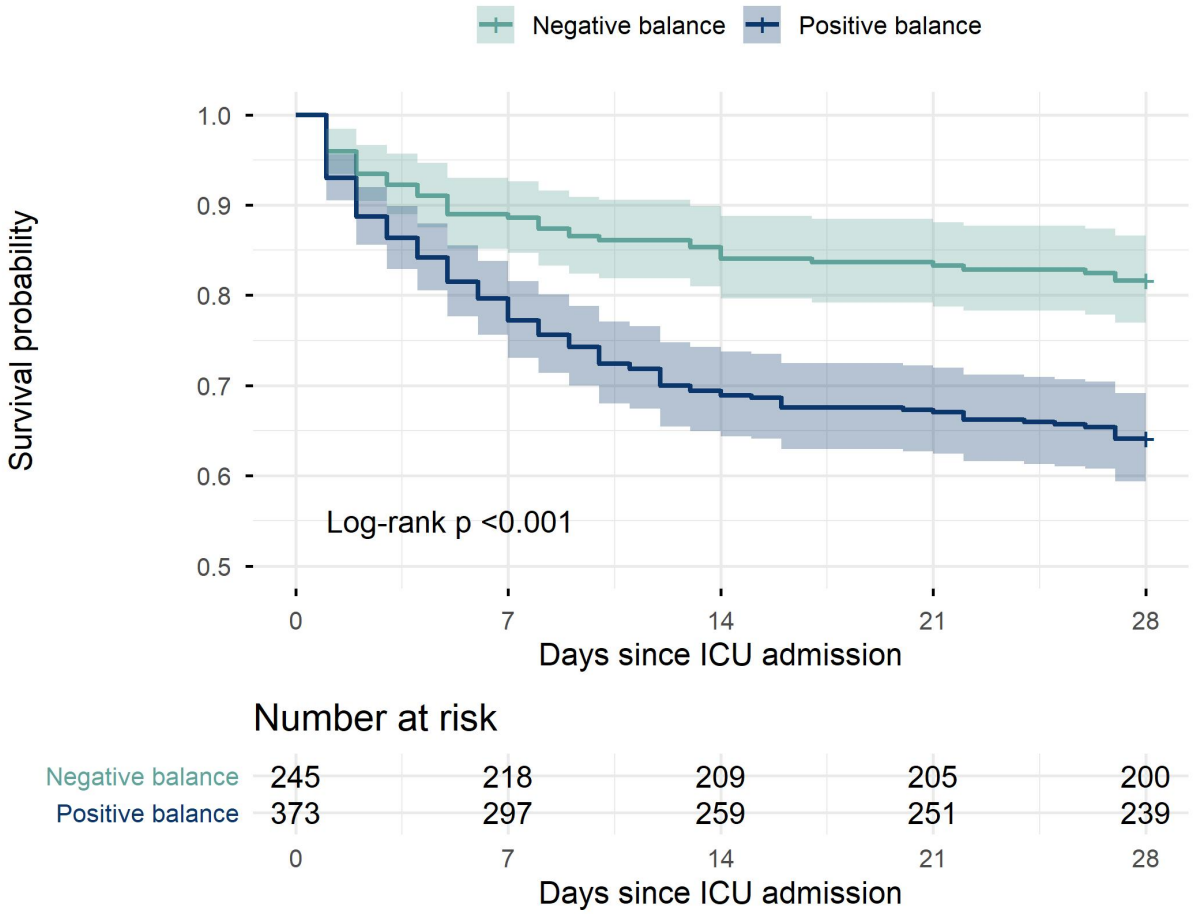


Figure 1. Kaplan-Meier survival to day 28 by fluid-balance group, with numbers at risk

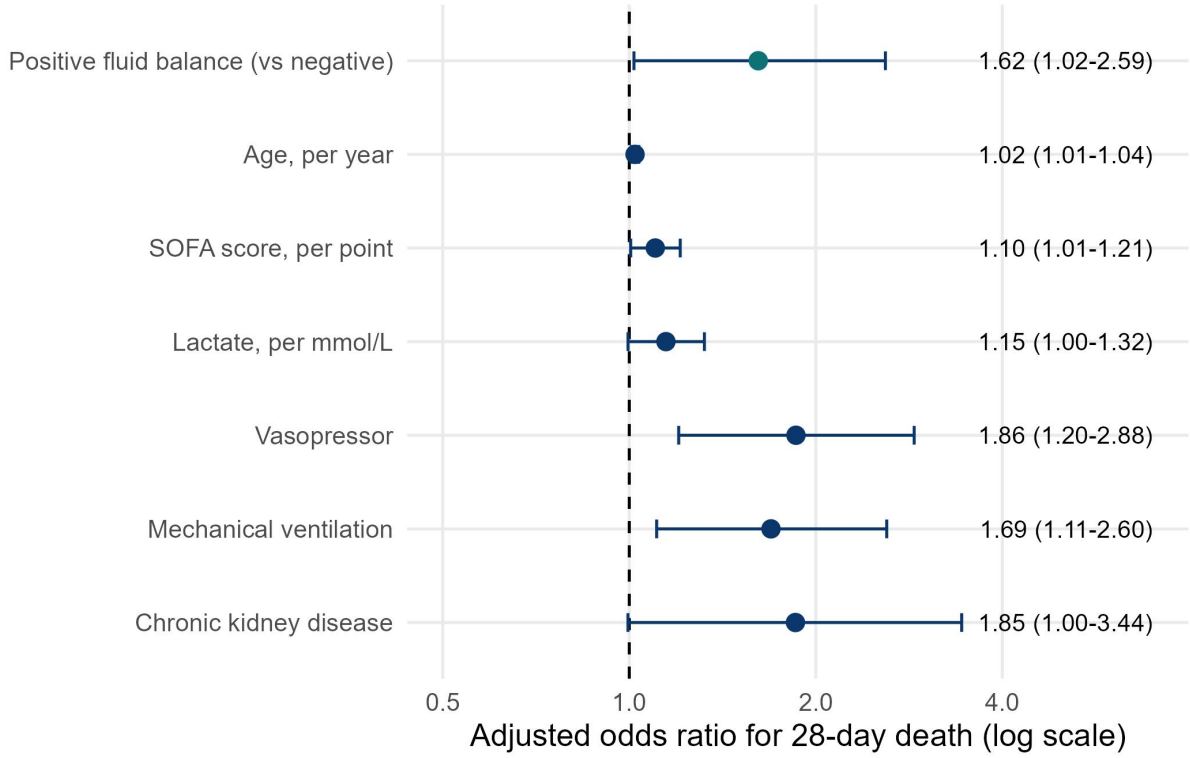


Figure 2. Adjusted odds ratios for 28-day death (forest plot)

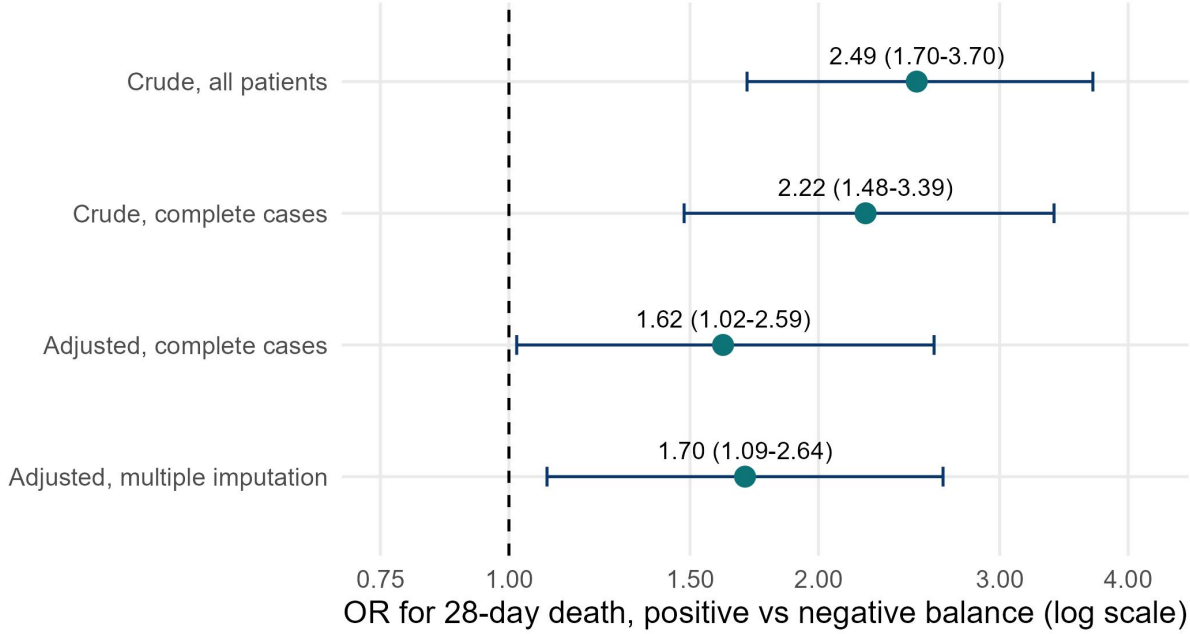


Figure 3. The odds ratio for positive balance before and after adjustment

5. Other analyses

Table 5. Pre-specified effect modification by heart failure

Subgroup	Deaths, negative balance	Deaths, positive balance	Adjusted OR (95% CI)	p for interaction
No heart failure	34/191 (18%)	108/330 (33%)	1.29 (0.77 to 2.19)	0.031
Heart failure	11/54 (20%)	26/43 (60%)	4.39 (1.65 to 12.38)	0.031

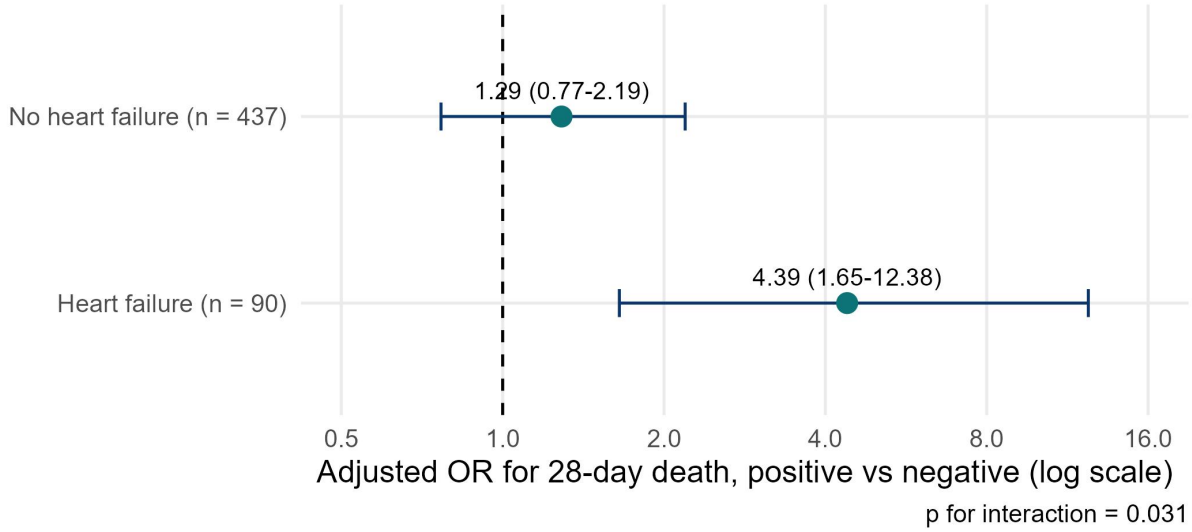


Figure 4. Odds ratio for positive balance in patients with and without heart failure

Table 6. Sensitivity analyses for the primary association

Analysis	n	OR (95% CI)	p
Crude, all patients	618	2.49 (1.70 to 3.70)	<0.001
Crude, complete cases	527	2.22 (1.48 to 3.39)	<0.001
Adjusted, complete cases (main)	527	1.62 (1.02 to 2.59)	0.044
Adjusted, multiple imputation (m = 20)	620	1.70 (1.09 to 2.64)	0.020

6. What this means (plain language)

• Patients with a positive fluid balance died more often (36% vs 18%). But they were also sicker.
• After taking severity into account, the association became much smaller (OR fell from 2.49 to 1.62). Much of the crude difference was confounding by indication.
• A modest association remained: the 95% CI only just excludes 1, and the Cox model gives a similar HR whose CI just touches 1. The evidence is weak-to-moderate, not strong.
• The association looked stronger in patients with heart failure (p for interaction 0.031). This was pre-specified but is based on only 90 patients with heart failure in the adjusted model (complete cases), so treat it as a hypothesis.

7. What this does NOT mean

• It does NOT show that fluid CAUSES death. This is an observational study: write 'associated with', not 'caused' or 'increased'.
• It does NOT tell clinicians to restrict fluid. Only a randomised trial (like Paper 1) can test a fluid strategy.
• A p-value of 0.044 is not 'proof'; a CI from 1.02 to 2.59 is compatible with a tiny or a large effect.

8. Limitations to discuss (hints)

• Residual confounding: SOFA and lactate measure severity imperfectly; unmeasured severity may explain the remaining association.
• Reverse causation: patients who do badly (shock, oliguria) accumulate fluid because they are dying, not the other way round.
• Fluid balance is measured over days 1-3; patients who died before day 3 had less time to accumulate fluid (immortal-time issue).
• Missing lactate (15%): the complete-case model lost 91 patients; multiple imputation gave a similar answer (Table 6).

- Insensible losses are not recorded; the dichotomy positive/negative throws away information (see Q2 for the continuous version).

Which STROBE items this pack helps you write

- Item 4-5: design and setting (section 2)
- Item 6: participants and eligibility
- Item 7: variables (exposure, outcome, confounders and the effect modifier, heart failure)
- Item 12a-e: statistical methods incl. confounding, interaction, missing data, sensitivity analyses
- Item 13: participant numbers (flow sentence)
- Item 14: descriptive data (Table 1, missing data)
- Item 15: outcome data (Table 2)
- Item 16: unadjusted and adjusted estimates with 95% CI (Tables 3-4)
- Item 17: other analyses (subgroup/interaction and sensitivity, Tables 5-6)